Rohit Das

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Summary

AI/ML Researcher with a Master's degree in Computer Science (specializing in 3D Computer Vision and AI). Experience in delivering projects in generative AI, 3D vision, AR/VR and applied ML in research labs and startups. Proven ability to translate research into real-world solutions such as virtual try-on, Avatar creation, 3D lip-sync, and texture generation.

Work Experience

Research Consultant at Sony Research India, India Sep 2025 - Present

- Working on State of the Art 3D-lipsync techniques for Sony Entertainment Limited
- Created a pipeline which provided the best facial region editing than any product in the market
- Currently working on SOTA rendering engine for AI-lipsync

  Frameworks: Gaussian Splats, NeRF, 3DMM, Neural Rendering

AI Engineer(Consultant) Jan 2024 - Dec-2025

Worked with multiple companies as a freelance consultant.

- (Stealth Startup, Tripura) Creating a Legal RAG saas system for Tripura judiciary system
  Frameworks: PyTorch, Ollama, VectorDB, AnythingLLM, Docker
- (Ctruh, Bangalore) Suggested masking strategy for Indian wear on V-TON system and designed a pipeline for 3D cloth deformation on 2D images using Fabric estimation and DrapeNet.
  Frameworks: PyTorch, CNN, U-net
- (Headspace, SF) Created an acne detection & removal pipeline for celebrity headshots.
  Frameworks: SAM, Grounding DINO, ControlNet, FastAPI, Docker
- (Estilo-AI, Ghana) Created a body-part segmentation for a V-TON system as well as implemented real-time rendering.
  Frameworks: SAM, Grounding DINO, Transformers, ControlNet, PyTorch, OpenCV, FastAPI, Docker
- (Global Digital Mojo, Taiwan) Created a survey on explainable AI usage and its future implications.
  Frameworks: SHAP, LIME

AI Engineer(Intern) at Bifrost, Singapore Sep 2023 - Feb 2024

- Reduced text-to-texture generation time from 45 minutes → 90 seconds.
  Frameworks: Blender, Stable Diffusion, ControlNet, Pytorch3D

Junior Researcher at CI3D Lab (NTUST, Taiwan) 2021 - 2023

- Researched and developed a novel method for 3D face texture estimation from single 2D image using StyleGAN3 and 3D Morphable Models(3DMM).
  Frameworks: PyTorch3D, Open3D,OpenGL,Blender,Pytorch,OpenCV, Docker

Education

2021 - 2023 M.S in Computer Science from **NTNU, Taiwan**
2014 - 2018 B.Tech in Computer Science from **CIT, Kolkata (India)**

Independent Projects & Publications

3D-GANTex GitHub

- Used StyleGAN3 + 3DDFA-V2 to reconstruct 3D face models from single images with consistent multi-view textures.
- Achieved PSNR **28.7** / SSIM **0.91** vs baselines on FFHQ dataset.
  Frameworks: PyTorch3D, OpenCV, Pytorch

Mesh from Video GitHub

- Designed a novel pipeline for 3D face texture estimation using 3DDFA-V3.
- Explored limitations due to dependence on FLAME coefficients.
  Frameworks: PyTorch3D, OpenCV, Pytorch, FLAME, 3DMM, 3DDFA

Publication: A Survey of the Normal Map Generator of GIMP from Single Shot Human Face Image. 3DSA-2022 ResearchGate

- The study explores the generation of near-accurate normal maps from a single image of a human face using the FFHQ dataset and GIMP.
- The research demonstrated effective methods for producing normal maps, contributing to advancements in texture mapping technologies.

Publication: 3D Face Reconstruction with StyleGAN3-based Multi-View Images and 3DDFA based Mesh Generation Arxiv

- A novel method utilizing StyleGAN and 3DMM to generate front face mesh from single 2D pose image.
- The pipeline achieved near to accurate texture from a single 2D mesh and helps in predicting unseen regions of the face.

Skills

Programming & ML: Python, C++, PyTorch, TensorFlow, CUDA, Transformers
Computer Vision & 3D: YOLO, DINO, NeRF, 3DDFA, FLAME, COLMAP, OpenCV, Open3D
Deployment & MLOps: Docker, FastAPI, ONNX, TensorRT, AWS